

Project: Clinical Notes Topic Modeling (gensim LDA) + Downstream Prediction

Goal

1. **Unsupervised topic discovery** from short clinical notes using **gensim LDA**.
2. Turn per-document **topic proportions** into features and compare downstream models:
 - **Logistic Regression** on TF-IDF
 - **Logistic Regression** on **LDA topic features**
 - **(Stretch)** Sentence-embedding features (e.g., MiniLM / BioClinicalBERT)
Outcome can be a small synthetic label (e.g., “deterioration flag”) or a proxy label (e.g., “needs ICU review”).

Dataset (medium, synthetic or public)

- **Option A (Synthetic; recommended for class):** Generate ~8,000 short notes with patterns (stable vs deteriorating language) + a binary label `needs_icu_review` (0/1).
- **Option B (Public):** Use a small subset of **MIMIC-III/IV NOTEEVENTS** (requires credentialed access). Keep ~10–20k notes; derive a weak label (e.g., whether note contains any “pressors/intubation/lactate>2.0” keyword).

Dataset generation and starter code

Provided separately

What to Compare & Report

- **Modeling paradigms:** TF-IDF vs **LDA topic proportions** vs **sentence embeddings** for predicting `needs_icu_review`.
- **Topic interpretability:** show top words per topic; assign human-readable labels (e.g., “hemodynamic instability”).
- **Ablations:** vary **k** (5, 7, 10, 12), **stop-word lists**, lemmatization on/off, and `no_below/no_above`.
- **Metrics:** ROC-AUC (primary), PR-AUC, macro-F1 at a chosen threshold.
- **Qualitative:** examples where LDA helps vs where it fails (short, noisy notes).

Stretch: Try a healthcare model like emilyalsentzer/Bio_ClinicalBERT via sentence-transformers or HF to do domain trained models comparison.)

Evaluation rubric (100 pts)

- Translate business problem to an optimal technical solution, including approach, options explored, logical sequencing of the workflow and tasks involved: **20**

- Data & Preprocessing (tokenization, stop words, lemmatization, dictionary/corpus) : **10**
- LDA modeling & topic quality (coherence, topic interpretability, k selection): **20**
- Downstream comparison (TF-IDF vs LDA vs embeddings) with solid metrics & fair splits: **10**
- Analysis & Insights (when topics help, error cases, healthcare relevance/ethics) – **10**
- Presentation clarity: **10**
- Teamwork: **10**
- Timely completion and submission: **10**

Submission Checklist

- make_notes.py, lda_train.py, lda_features.py, downstream_compare.py, (optional) lda_viz.py
- requirements.txt and a short **README** with exact reproduce steps
- Metrics table + 1–2 plots (ROC/PR) + a one-page write-up labeling topics and discussing findings